

Question

Known: For the diprotic acid H_2CO_3 , $K_{a1} = 4.2 \times 10^{-7}$, $K_{a2} = 5.6 \times 10^{-11}$, and for $CaCO_3$, $K_{sp} = 2.9 \times 10^{-9}$. Calculate the solubility of $CaCO_3$ in pure water and in 1L of 0.01 mol/L $NaHCO_3$ solution (in mol/L), respectively. The following options provide possible answers for the two solubilities, with the units already given in the question. Choose the correct option (an error within 15% is considered correct):

A. 5×10^{-5} , 1.8×10^{-5}

B. 5×10^{-5} , 2.5×10^{-5}

C. 5×10^{-5} , 3.3×10^{-5}

D. 7×10^{-5} , 1.8×10^{-5}

E. 7×10^{-5} , 2.5×10^{-5}

F. 7×10^{-5} , 3.3×10^{-5}

G. 9×10^{-5} , 1.8×10^{-5}

H. 9×10^{-5} , 2.5×10^{-5}

I. 9×10^{-5} , 3.3×10^{-5}

J. All other options are incorrect.

Answer

Correct Answer: **H**

Detailed Explanation

In pure water:

$CaCO_3 \rightarrow Ca^{2+} + CO_3^{2-}$, Let the solubility of $CaCO_3$ be x mol/L, then the total concentration of Ca^{2+} and H_2CO_3 in the solution are both x mol/L. The concentration of CO_3^{2-} can be expressed using the distribution coefficient. Therefore, according to the precipitation-dissolution equilibrium:

$$[Ca^{2+}][CO_3^{2-}] = K_{sp}$$

CHECKPOINT

1 PTS

Precipitation-dissolution equilibrium: $[Ca^{2+}][CO_3^{2-}] = K_{sp}$

$$x^2 \cdot \frac{K_{a1}K_{a2}}{[H^+]^2 + K_{a1}[H^+] + K_{a1}K_{a2}} = K_{sp}$$

In addition, according to charge conservation, we can list:

$$[H^+] + 2[Ca^{2+}] = 2[CO_3^{2-}] + [HCO_3^-] + [OH^-]$$

CHECKPOINT

1 PTS

Charge conservation: $[H^+] + 2[Ca^{2+}] = 2[CO_3^{2-}] + [HCO_3^-] + [OH^-]$

$$[H^+] + 2x = x \cdot \frac{2K_{a1}K_{a2} + K_{a1}[H^+]}{[H^+]^2 + K_{a1}[H^+] + K_{a1}K_{a2}} + [OH^-]$$

Solving the simultaneous equations yields $x = 1.0 \times 10^{-4} \text{ mol/L}$

CHECKPOINT

1 PTS

The solubility of $CaCO_3$ in pure water is $1.0 \times 10^{-4} \text{ mol/L}$

In $NaHCO_3$ solution:

$CaCO_3 \rightarrow Ca^{2+} + CO_3^{2-}$, Let the solubility of $CaCO_3$ be $x \text{ mol/L}$, then the total concentration of Ca^{2+} in the solution is $x \text{ mol/L}$, and the total concentration of H_2CO_3 is $(x+0.01) \text{ mol/L}$. The concentration of CO_3^{2-} can be expressed using the distribution coefficient. Therefore, according to the precipitation-dissolution equilibrium:

$$[Ca^{2+}][CO_3^{2-}] = K_{sp}$$

$$x(x + 0.01) \cdot \frac{K_{a1}K_{a2}}{[H^+]^2 + K_{a1}[H^+] + K_{a1}K_{a2}} = K_{sp}$$

In addition, according to charge conservation, we can list another equation:

$$[H^+] + 2[Ca^{2+}] + [Na^+] = 2[CO_3^{2-}] + [HCO_3^-] + [OH^-]$$

CHECKPOINT

1 PTS

Charge conservation: $[H^+] + 2[Ca^{2+}] + [Na^+] = 2[CO_3^{2-}] + [HCO_3^-] + [OH^-]$

$$[H^+] + 2x + 0.01 = (x + 0.01) \cdot \frac{2K_{a1}K_{a2} + K_{a1}[H^+]}{[H^+]^2 + K_{a1}[H^+] + K_{a1}K_{a2}} + [OH^-]$$

Solving the simultaneous equations yields $x = 2.3 \times 10^{-5} \text{ mol/L}$

CHECKPOINT

1 PTS

The solubility of $CaCO_3$ in 1L 0.01 mol/L $NaHCO_3$ solution is $2.3 * 10^{-5} mol/L$

Therefore, choose option H